

Classical machine learning and deep neural network ensemble model for GPS-based activity recognition

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ABSTRACT

Our KDDI Research team proposes an ensemble model for the Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge using GPS data. During preprocessing, we corrected the GPS dataset that contains errors and missing values using a Kalman smoother. Since this smoother can be processed offline, it can be used to correct backward using the recorded future locations in the dataset. Further, by using features that have similar distributions in the train subject's data and the other subjects' datasets, we could achieve robust feature selection across multiple subjects. The first stage of our ensemble model employs classical machine learning and deep neural network approaches independently, specifically, LightGBM and LSTM, respectively. The second stage calculates the weighted average of the outputs of both approaches. Our results show the improved accuracy contributed by our ensemble, suggesting that it effectively makes use of both statistical and non-statistical features given the suitable base models. We confirmed that the use of Kalman-smoother, selection of features with similar distributions across subjects and ensemble modeling contributed to improving the accuracy of both train and validation datasets.

KEYWORDS

Locomotion activity recognition; deep neural networks; lightgbm; Kalman-smoother

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1 INTRODUCTION

Locomotion activity recognition from GPS, WiFi and cell tower scans is a very challenging task. The benefits, however, of successfully performing this task has been shown to be far-reaching. For instance, by being able to recognize locomotion changes, the activities of a person can be inferred, and the resulting data can later be used to promote health and exercise habits. It can also be used for real-time notifications and promoted related advertisements.

Due to the proliferation of the now ubiquitous smartphones (and other smart devices), research on recognizing locomotion activity primarily from the use of motion sensors has progressed. This phenomenon has coincided with the surge of deep neural network approaches, which in contrast to traditional machine learning models have been enabling efficient learning from huge volumes of data. A major difficulty, however, when using motion sensors is that the orientation of the smartphone significantly differs depending on the subject, and the locations where user data were collected may also vary. Thus, it is easy for noise to accompany the large volume of data taken from various subjects and in various locations.

It is against the above backdrop that the utility of the data used in the Sussex-Huawei Locomotion-Transportation (SHL) recognition challenge, specifically, GPS, WiFi and cell tower scans data, increases since compared to motion sensors these are less dependent to, and not severely influenced by, subjects' varying behaviors. It is likely that more robust models can be inferred from using such data. A major challenge, however, is that the receptiveness of the device to radio wave signals can become unstable, for instance, being underground, deep in the mountains or in some obscure place.

Our KDDI Research team aims to address the challenges we outlined above through our proposed three-stage ensemble modeling, i.e., offline GPS data correction is first performed during the preprocessing stage using a Kalman-smoother, features with similar distributions in the train and other datasets are then selected, and in the last stage, the activity recognition model is learned using an ensemble of classical and deep neural network models.

We structure the paper as follows. Section 2 lays down the background and related works, while Section 3 characterizes the datasets used to infer our models together with their accompanying challenges. Section 4 then describes the preprocessing and feature engineering tasks, while Section 5 elucidates our ensemble model. We then discuss our experimental results and analyses in Section 6, and finally, we conclude in Section 7.

2 BACKGROUND AND RELATED WORKS

Human activity recognition (HAR) has been extensively explored and investigated, evidenced by the plethora of surveys that have been published on this subject (e.g., [17], [26], [9] and [12], among others), covering various domains, applications and methods employed, and have proved to achieve promising results. The kinds of data that are being handled by researchers in this area vary from common activities such as locomotion to specialized ones like nurse care [14]. This said, the scope of HAR research is still expanding thanks to the rapid and continuous development of various sensor-based smart and IoT devices. The proliferation of these easily accessible and easy to use sensor-based devices consequently paved the way for significantly increasing the possibility of obtaining efficiently massive amount of human behavior data. What has eventually emerged together with this are more advanced machine learning algorithms that have made remarkable progress for HAR research. [5][18]. Inspired by this development, the SHL recognition challenge has been held for four years in a row, promoting the use of public datasets to benchmark various HAR approaches [3].

In the previous three years of competition, data obtained from the use of motion sensors (accelerator and gyro, etc.) were used [24][20][21]. Moreover, traditional machine learning and deep neural network models were proposed [23][7] to tackle the challenges arising from learning HAR models through the use of motion sensors. In particular, it has been reported that an ensemble model contributed to improving accuracy [15]. This year's SHL recognition challenge is relatively new in that it uses GPS, WiFi, and cell tower scan data on top of motion sensor data.

3 CHALLENGE AND DATASET

The goal of the SHL recognition challenge is to recognize eight modes of human locomotion and transportation. The SHL dataset consists of the modes as labels and the time-series data of GPS reception and location, WiFi reception and GSM cell tower scans [10][22]. Table 1 shows the number of labels and the users included in the train, validation and test datasets. The difficulty here is that the user in the dataset used for training the models is different from the other two users in the validation and test datasets, with user distinction not explicitly stated (i.e., the validation and test datasets contain no user column, and thus, it cannot be known if samples belong to User 2 or User 3).

Figure 1 shows the histogram of the labels in the train and validation datasets. What can be observed is that the distributions of the labels are different. Although this is a necessary condition for creating a more generalized model, it is a very challenging task considering the model could easily overfit across users. Further, Table 2 shows the input sensor data has four modalities, namely, GPS reception and location, WiFi reception and cell tower scans. All are time-series data and require suitable feature engineering based on domain knowledge.

In addition, these sensor data present two issues, namely, portions of the sensor data are often missing due to poor satellite reception, hardware glitches or human errors, among others, and sampling rate is asynchronous, albeit assumed to be roughly 1Hz.

Table 1: Number of labels in the SHL challenge dataset

	#Labels	User
Train	980527	User 1
Validation	143958	Users 2, 3
Test	671172	Users 2, 3

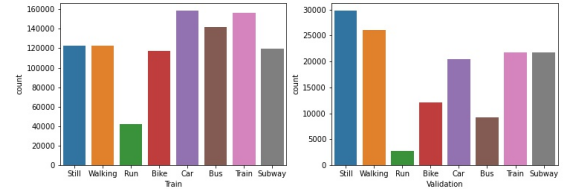


Figure 1: Distributions of train and validation labels

4 PREPROCESSING AND FEATURE ENGINEERING

We divided the entire labeled data that has a whole time-series duration T into disjoint sub-time series, $\tau_1, \tau_2, \dots, \tau_T$, with each sub-time series τ separated by two seconds or more. Each of the processes described below is carried out for each sub-time series.

4.1 Preprocessing

The preprocessing pipeline is as follows:

- align the asynchronous time stamps of the sensor data to match those of the data labels,
- correct the latitude and longitude of the GPS location data per second using a Kalman-smoother,
- calculate the velocity from the latitude and longitude of the GPS location data, at the same time removing as outliers those over 200km/h, and
- linearly interpolate for the missing values.

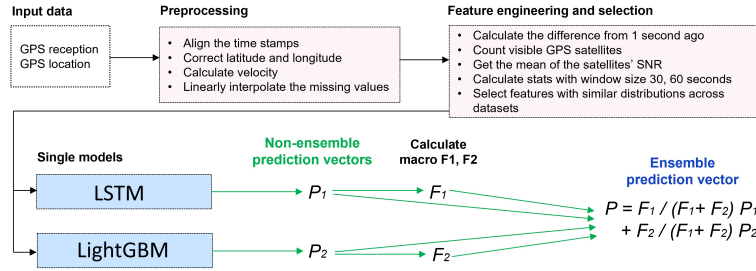
A Kalman-smoother is an extension of the Kalman-filter. Although the Kalman-filter predicts and updates the present time series using past data, the Kalman-smoother corrects the prediction results of the Kalman-filter backforward from future data. It has been reported that better results could be obtained by correcting inaccurate GPS data [25][16][8].

4.2 Feature engineering

From the GPS position data, we computed for the velocity, as well as the variations in altitude and accuracy one second prior. For instance, if the altitude (accuracy) one second ago was 1000 (50) meters and the current altitude (accuracy) is 990 (100) meters, the variation in altitude (accuracy) is -10 (50) meters. Second, we counted from the GPS reception data the number of visible GPS satellites and computed for the mean of those satellites' SNR, which could indicate the instability of the satellites (e.g., when the user is in the subway). Afterwards, we created the different features that were used by our traditional machine learning and the deep neural network models. Specifically, from the extracted GPS position and reception data features, we computed the statistical values (mean,

Table 2: Sensor data format for this challenge

Modality	Data format
GPS reception	Epoch time[ms], satellites ID, SNR, Azimuth[degrees], Elevation[degrees]
GPS location	Epoch time[ms], Accuracy of location[m], Latitude[degrees], Longitude[degrees], Altitude[m]
WiFi reception	Epoch time[ms], BSSID, SSID, Frequency[MHz], Capabilities
Cell tower	Epoch time[ms], type of cell(LTE, GSM, WCDMA), isRegistered flag, Area code and signal level for each type of cell, etc.

**Figure 2: Proposed model**

standard deviation, max and min) using window sizes of 30 and 60 seconds, which are then used by the traditional machine learning model. On the other hand, we used for the neural network model the raw GPS position and reception data features using also window sizes of 30 and 60 seconds.

4.3 Feature selection

Table 1 shows that the subject of the train data is User 1, while the subjects of the validation and test data are User 2 and User 3, respectively. With this set-up, it is highly possible that our model would overfit with the unique features of User 1. To evade this, we use a technique that selects features that are similar in distribution and are user-independent across the datasets [1][2], as follows:

- (1) add a flag column “train or not” to the datasets to indicate whether it is a train data or not, that is, the column is filled with 1 for the train dataset and 0 for the other datasets;
- (2) concatenate the train and other datasets;
- (3) train Random Forest (RF) [6] model to classify the “train or not” and calculate the feature importance and AUC;
- (4) remove the feature with the highest feature importance; and
- (5) repeat steps (3) and (4) until the AUC is less than 0.6.

The AUC value 0.6 is a commonly used empirically-derived threshold. Experimentally, for example, there will be no features left to learn with if the AUC threshold is 0.55. We have yet to investigate, however, if an AUC of 0.65 or higher will yield better results.

Since our RF is a classification model of whether it is a train dataset or not, the AUC increases as the distributions of the features in the train and other datasets differ. Conversely, if the AUC deteriorates to some extent from removing the features that are important for classifying the datasets, it means that the remaining features do not contribute to the classification task and are deemed similar across datasets. The features that were selected by our five-step algorithm above include the following:

- features from window size 30: standard deviation of velocity, mean number of visible GPS, mean SNR, and max SNR; and
- features from window size 60: mean velocity, max velocity, min velocity, mean variation of altitude, min variation of altitude, max variation of accuracy, standard deviation of the variations in accuracy, mean number of visible GPS, standard deviation of the number of visible GPS, mean SNR, max SNR, and min SNR.

These serve as inputs to our model, which we elucidate next.

5 PROPOSED MODEL

We created an ensemble model based on both traditional machine learning, which are the Random Forest (RF) [6] and LightGBM [13], and deep neural network, i.e., LSTM [11], models. While we use basic statistical features for RF and LightGBM, we use the raw features for LSTM (see previous sections). The hyperparameters of these models are optimized by Optuna [4], which are shown below.

- Random Forest : n_estimators = 250;
- LightGBM : lambda_l1 = 0, lambda_l2 = 0, num_leaves = 9, feature_fraction = 0.7, bagging_fraction = 1.0, bagging_freq = 0, min_child_samples = 20; and
- LSTM : hidden_size = 250, num_of_layers = 2.

The weighted average ensemble method of the top two models with high macro F1 scores is then used (Figure 2, for instance, shows that the top two models are LSTM and LightGBM). That is, if the prediction probability vector of model $i (= 1, 2, 3)$ is p_i and its macro F1 score is F_i , and assuming $F_1 > F_2 > F_3$, then the prediction probability vector of the ensemble model is $P = \frac{F_1}{(F_1+F_2)}p_1 + \frac{F_2}{(F_1+F_2)}p_2$, and the label corresponding to the element that maximizes this vector P is used as the final recognition label.

6 RESULTS AND DISCUSSION

Firstly, Tables 3 and 4 show the results of our experimenting with the three base models and with the train and validation datasets of

this challenge as inputs. We ran our experiments in a TITAN V GPU machine with an Intel(R) Core(TM) i7-9800X CPU at 3.80GHz and 16GB RAM. The training time lasted for roughly 27 hours, and the testing time roughly 30 minutes (note, however, that the results of our testing are instead shown in the summary report of this year’s SHL challenge). When the base models are used with both Kalman-smoother and feature selection (second column of the tables), higher macro F1 score is achieved for both train and validation datasets. Further, we can confirm that the GPS location information has been corrected successfully, and the invariant features can be selected effectively for the train and other datasets.

Our results show LSTM scores the highest during validation, followed by LightGBM. Thus we ensembled LSTM and LightGBM, with the results of this ensemble shown in Table 5 and Figure 3. The macro F1 score improved with this ensemble approach (Table 5). However, the confusion matrix (Figure 3) shows “Still” and “Train” are often mispredicted as “Subway”, while the converse does not hold, i.e., “Subway” is seldomly mispredicted as “Still” or “Train”. Although we have created features based on the number of visible GPS satellites and the mean of those satellites’ SNR, which are indicative of instability, to infer the satellites that are underground, the degree of instability in “Still” and “Train” is still the same. What we think is a way around this problem, which is worth exploring in the future, is to create some unique features for “Subway” from the WiFi reception and cell tower scan data, both of which are available in this challenge dataset anyway.

The optimal weights we obtained for the LSTM and LightGBM ensemble were $w_1 = \frac{0.867}{(0.867+0.824)} = 0.513$ and $w_2 = \frac{0.824}{(0.867+0.824)} = 0.487$, respectively. In addition, Table 6 shows the results we obtained when changing the weight w_1 , increasing it each time by 0.1. The most accurate is when $w_1 = 0.5$, but slightly less than the optimal weight we obtained.

We also experimented the case that the top two models are the RF and LGBM. Thus, both are traditional machine learning models. The performance, however, significantly deteriorated (we obtained macro F1 equal to 0.701 during validation). Thus, we posit the accuracy of our ensemble model can be improved only by using models that have different characteristics, such as the Tree (RF or LightGBM) and DNN (LSTM) models. Further, we have tried just three models, but with our top-two ensemble approach already resulting to sufficiently high macro F1. When we tried the ensemble of the three models, the validation result deteriorated to a macro F1 of 0.787 with the weights all equal to 1/3. We posit that an ensemble of several base models that proved highly predictive when independent may accelerate overfitting during validation. Lastly, it would be worthwhile in the future to investigate further not only the ensemble of the best two models possessing different characteristics, but also all the pairwise combinations of different models whether or not their characteristics are the same.

7 CONCLUSION

We proposed a model that recognizes eight human locomotion activity using GPS data within the Sussex-Huawei Locomotion-Transportation recognition challenge. Our model is an ensemble model based on traditional machine learning and deep neural network, which demonstrated higher macro F1 score than compared to

Table 3: Base models’ macro F1 scores for the train dataset

Base model	Has both	w/o Kalman-smoother	w/o feature selection
Random Forest	0.790	0.641	0.744
LightGBM	0.886	0.850	0.801
LSTM	0.890	0.822	0.834

Table 4: Base models’ macro F1 scores for the validation dataset

Base model	Has both	w/o kalman-smoother	w/o feature selection
Random Forest	0.688	0.602	0.552
LightGBM	0.824	0.720	0.602
LSTM	0.867	0.797	0.670

Table 5: Ensemble model score

Dataset	macro F1	macro precision	macro recall
train	0.901	0.898	0.943
validation	0.887	0.875	0.933

Table 6: macro F1 scores from the weighted average strategy for the validation dataset

w_1	0	0.1	0.2	0.3	0.4	0.5
macro F1	0.824	0.850	0.859	0.871	0.880	0.886
w_1	0.6	0.7	0.8	0.9	1	
macro F1	0.883	0.880	0.869	0.870	0.867	

just using the base models independently. Furthermore, the macro F1 score is improved by correcting the GPS location with a Kalman-smoother during preprocessing and then selecting the features that manifested similar distributions in the train and other datasets. In particular, the macro F1 score for the validation dataset greatly deteriorated in the absence of the feature selection stage.

However, there remains the non-trivial issue of misprediction (such as mispredicting “Subway” as “Still” or “Train”). We assume that this is due to the unstable state of the available GPS satellites other than when the user is in the “Subway”, and we hypothesize that this problem can be resolved by using WiFi reception and cell tower reception data, which were not used at this time.

Since our proposed model is able to achieve high macro F1 score even for the validation dataset that has a different distribution to that of the train dataset, it can be considered a robust model. We expect that even with the test dataset whose label distribution is unknown our model’s performance may result to high macro F1 score. The recognition result for the testing dataset is presented in the summary paper of this challenge [19].



Figure 3: Confusion matrix of ensemble model for validation dataset

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